

Element C

In the design process for a new animal lifting device for the Nashville Zoo Veterinarian Hospital, it is crucial to determine the design requirements based on prior research and designs. Based on our research in element A and the needs of our primary stakeholders we concluded that our criteria includes; time to load, weight bearing capacity (50-1100 pounds), durability (no critical failures during use), ease of use (number of steps from loading to lift), balance (ability to balance weight 50/50 across its center of mass for a period of time), removability, and ensuring the device does not hurt the animals more than they already are.

When creating a comprehensive list of requirements, we need to prioritize them. This prioritization is crucial because it allows us to focus our efforts on the most critical aspects of the design. Here is our list from highest to lowest priority;

1. weight bearing capacity (50-1100 pounds)
2. balance (ability to balance weight 50/50 across its center of mass for a period of time)
3. durability (no critical failures during use)
4. ensuring the device does not hurt the animals more than they already are.
5. removability
6. ease of use (number of steps from loading to lift)
7. time to load

The majority of our requirements are high priority because they are the fundamental elements that must be met for the device to be successful and used long term at the hospital. Most of our criteria is measurable, as it makes it easier for us to ensure we have met the criteria but others such as “harming the animals” will need to be determined not by measuring but by physical testing.

Using a decision matrix and extensive discussion where we consult our criteria, we continue to refine our design requirements to ensure a tangible and viable solution. Based on past designs we concluded that balance, weight capacity, and durability should be our main focus when creating the lift. Therefore finding a stronger material (superstrut) and a balance of some kind (engine lift balance) can be the foundation of our design that allows us to build our solution.

The design requirements for the animal lift project are also carefully chosen in order to meet the specific needs of the primary stakeholders, which include the zoo animals in need of treatment, Dr. Steve Wancea, and his assistants. The weight-bearing capacity requirement ensures that the device can safely lift a wide range of potential animals in need. This requirement goes hand in hand with the balance requirement, because it enables the lift to balance weight evenly across its center of mass, to ensure the safety and stability of the animals during lifting procedures. Importantly, the requirement to ensure the device does not hurt the animals more than they already are highlights a deep commitment to animal welfare, a crucial aspect for Dr. Wancea and his team, who prioritize the well-being of the animals in their care. The final requirement that was not executed efficiently last time, durability. This is a crucial requirement because it mandates no critical failures during use, guarantees the device's reliability and long-term functionality, and is essential for Dr. Wancea and his team's operations in the long run. Overall, these design requirements are meticulously crafted to meet the needs of the primary stakeholders, ensuring safe, efficient, and humane animal lifting procedures under Dr. Wancea's supervision and approval.